THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 44 cm	$2\pi r = 2 \times \frac{22}{7} \times 7 = 44$
2	**(d)** 220 cm	$2\pi r = 2 \times \frac{22}{7} \times 35 = 220$
3	**(b)** 10.5 cm	$r = \frac{C}{2\pi} = \frac{66 \times 7}{44} = 10.5$
4	**(b)** 11 cm	$\frac{90}{360} \times 2\pi r = 11$
5	**(a)** 30 cm^2	$s = 15$, area $= \sqrt{15 \cdot 10 \cdot 3 \cdot 2} = 30$
6	**(d)** 36 cm^2	$s = 18$, area $= \sqrt{18 \cdot 9 \cdot 8 \cdot 1} = 36$
7	**(d)** 35 cm^2	area $= \frac{1}{2}bh$
8	**(c)** 25 sq. units	$s = 10$; $\sqrt{(s-a)(s-b)(s-c)(s-d)} = 25$
9	**(b)** 1386 cm^2	$\pi r^2 = \frac{22}{7} \times 21^2 = 1386$
10	**(b)** πr	half of $2\pi r$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2\pi r = 44 \Rightarrow r = \frac{44 \times 7}{2 \times 22} = 7$ cm.	1+1
12	$2 \times \frac{22}{7} \times 10 = 62.857\ldots \approx 62.9$ cm.	1+1
13	$\frac{60}{360} \times 2 \times \frac{22}{7} \times 3.5 = \frac{11}{3} \approx 3.67$ cm.	1+1
14	Arc $= \frac{75}{360} \times 2 \times \frac{22}{7} \times 14 = \frac{55}{3}$ cm; perimeter $= \frac{55}{3} + 14 + 14 = \frac{139}{3} \approx 46.3$ cm.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Straights: $2 \times 84.39 = 168.78$ m (1). Two semicircles make one circle: $2\pi r = 2 \times 3.1416 \times 36.5 = 229.34$ m (1). Total ≈ 398.12 m (1).	1+1+1
16	Radius increases by 1.2 m (1). Two semicircles: extra distance $= 2\pi \times 1.2 = 2 \times 3.14 \times 1.2$ (1) $= 7.536 \approx 7.54$ m (1).	1+1+1
17	Triangle ABC (the centres and an intersection point) is equilateral, so $\angle CAD = 120^\circ$ (1). Each outer arc is $240^\circ = \frac{2}{3}$ of a circle (1). Perimeter $= 2 \times \frac{2}{3} \times 2\pi r = \frac{8\pi r}{3}$ (1).	1+1+1
	OR: Radii a', b', c', d'. Path $a = \pi a'$; path $b + c + d = \pi(b' + c' + d')$ (1). $PQ = 2a' = 2b' + 2c' + 2d' \Rightarrow a' = b' + c' + d'$ (1). Hence the lengths are equal (1).	1+1+1
18	Inscribed hexagon perimeter $= 6r < 2\pi r \Rightarrow \pi > 3$ (1). Circumscribed hexagon has side $\frac{2r}{\sqrt{3}}$ (from Pythagoras with half-side $\frac{r}{\sqrt{3}}$) so perimeter $= \frac{12r}{\sqrt{3}} = 4\sqrt{3}r > 2\pi r \Rightarrow \pi < 2\sqrt{3} \approx 3.46$ (2).	1+2
	OR: $s = 21$; area $= \sqrt{21 \times 8 \times 7 \times 6} = 84$ cm^2 (2). $\frac{1}{2} \times 14 \times h = 84 \Rightarrow h = 12$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cut the right triangle off one side and attach it to the other to form a rectangle of the same base b and height h (2). Rectangle area $= bh$ (1). For a thin parallelogram the foot of the perpendicular lies outside the base: shift triangles repeatedly (equal congruent triangles CDD', BAA') until the construction works (2).	2+1+2
	OR: Two congruent copies fit to make a parallelogram of base b and height h (2) ⇒ triangle $= \frac{1}{2}bh$ (1). Median AD: bases $BD = DC$, same height ⇒ equal areas (2).	2+1+2
20	(i) $36.5 + 0.3 = 36.8$ m (1). (ii) $168.78 + 2\pi(36.8) = 168.78 + 231.22 = 400$ m (2). (iii) Extra radius 1.22 m: $2\pi \times 1.22 \approx 7.67$ m (1). (iv) Each lane's radius increases by the same 1.22 m, so the extra bend distance is the same $2\pi \times 1.22$ for each pair (1).	1+2+1+1
	OR: (i) $s = a + b$; area $= \sqrt{(a+b)(b)(b)(a-b)} = b\sqrt{a^2 - b^2}$ (3). (ii) $h = \sqrt{25 - 9} = 4$, base 6: $\frac{1}{2} \times 6 \times 4 = 12$; formula: $3\sqrt{16} = 12$ ✓ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **Measuring Space: Perimeter and Area** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 88 cm	$2\pi r = 2 \times \frac{22}{7} \times 14 = 88$
2	**(c)** 7 cm	$r = \frac{C}{2\pi} = \frac{44 \times 7}{44} = 7$
3	**(c)** 22 cm	$\frac{60}{360} \times 2\pi r = 22$
4	**(d)** 11 cm	$\frac{45}{360} \times 2\pi r = 11$
5	**(c)** 84 cm^2	$s = 21$, area $= \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$
6	**(d)** 84 cm^2	area $= bh$
7	**(a)** 72 cm^2	area $= \frac{1}{2}bh$
8	**(b)** 60 sq. units	$s = 17$; $\sqrt{(s-a)(s-b)(s-c)(s-d)} = 60$
9	**(b)** 38.5 cm^2	$\pi r^2 = \frac{22}{7} \times 3.5^2 = 38.5$
10	**(d)** $\frac{\sqrt{3}}{4}a^2$	Heron with $s = \frac{3a}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumference $= \frac{22}{7} \times 56 = 176$ cm; distance $= 100 \times 176 = 17600$ cm $= 176$ m.	1+1
12	Perimeter $= 2\pi r$, so the ratio of perimeters equals the ratio of radii: $5 : 4$.	2
13	Arc $= \pi r = 22$ cm; diameter $= 14$ cm; perimeter $= 36$ cm.	1+1
14	$s = 12$; area $= \sqrt{12 \times 4 \times 4 \times 4} = \sqrt{768} = 16\sqrt{3}$ cm^2.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= 42$ cm^2 (Heron, $s = 21$) (2). $\frac{1}{2} \times 7 \times h = 42 \Rightarrow h = 12$ cm (1).	2+1
16	In $\triangle ABC$ let AD be the median, $BD = DC$ (1). $\triangle ABD$ and $\triangle ACD$ have the same height h from A (1). Areas $\frac{1}{2} \cdot BD \cdot h = \frac{1}{2} \cdot DC \cdot h$ — equal (1).	1+1+1
17	$s = \frac{3a}{2}$, $s - a = \frac{a}{2}$ (1). Area $= \sqrt{\frac{3a}{2} \cdot \frac{a}{2} \cdot \frac{a}{2} \cdot \frac{a}{2}}$ (1) $= \frac{\sqrt{3}}{4}a^2$ (1).	1+1+1
	OR: $s = 2a$; area $= \sqrt{(2a-a)^4} = \sqrt{a^4} = a^2$ (3), which is the known area of the square.	3
18	$s = 88$; $s - a = 63, 49, 36, 28$ (1). Area $= \sqrt{63 \times 49 \times 36 \times 28} = 1764$ cm^2 (2).	1+2
	OR: Circle $= \frac{22}{7} \times 21^2 = 1386$ cm^2 (1+1). Sector $= \frac{60}{360} \times 1386 = 231$ cm^2 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $s = a + b$; $\sqrt{b \cdot a \cdot b \cdot a} = ab$ ✓ (2). (ii) $s = 15$; area $= \sqrt{(15-6)(15-14)(15-5)(15-5)} = \sqrt{900} = 30$ (2). Height $= \sqrt{25 - 16} = 3$; $\frac{1}{2}(6 + 14) \times 3 = 30$ ✓ (1).	2+2+1
	OR: (i) $\frac{22}{7} \times 196 = 616$ m^2 (1). (ii) Outer radius 21 m; area $\frac{22}{7} \times 441 = 1386$ m^2 (2). (iii) $1386 - 616 = 770$ m^2 (1). (iv) $770 \times 50 = ₹38{,}500$ (1).	1+2+1+1
20	(i) $\frac{22}{7} \times 56 = 176$ cm (1). (ii) $\frac{176000}{176} = 1000$ (1). (iii) $500 \times 176 = 88000$ cm/min $= 880$ m/min $= 52.8$ km/h (2). (iv) Circumference 220 cm; ratio of distances $= 176 : 220 = 4 : 5$ (1).	1+1+2+1
	OR: (i) $s = 34$; area $= \sqrt{34 \times 17 \times 9 \times 8} = 204$ cm^2 (2). (ii) Altitudes $\frac{2 \times 204}{17} = 24$, $\frac{408}{25} = 16.32$, $\frac{408}{26} \approx 15.69$ cm (2). (iii) The altitude on the shortest side (24 cm) (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.